

TRANSSERIES / ASYMPTOTIC INVERSION / COUNTABLE EXPONENTIAL FEEDBACK

Beyond Finite-Action Folds

Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets

Research draft prepared for Vladimir Reshetnikov

29 September 2026

Abstract

Finite analytic action families generically produce square-root inverse singularities. A countable family can instead reach a nonanalytic boundary, where that conclusion fails. We study the explicit positive class

$$U_\beta(q) = \beta \sum_{j \geq 1} a_j q^j e^{j U_\beta(q)}, \quad a_j = a j^{-1-\alpha} \quad \text{outside a finite set,} \quad 1 < \alpha < 2.$$

At its critical coupling, we construct an actually convergent Hahn inverse with a finite positive generating grid, give a closed formula for every coefficient, and determine its exact finite-ramification criterion. An all-orders, uniformly additive large-sector expansion has coefficients expressed through derivatives of a stable density. A reflection identity identifies its terms with the Hahn coefficients and explains a precise arithmetic cancellation rule. We then prove a joint coupling–action–cutoff limit. The number of retained actions must exceed the stable scale $n^{1/\alpha}$ to preserve the n th coefficient asymptotically; at a comparable cutoff an explicit infinitely divisible density gives the exact limiting loss. Finally, we compute the critical-value and curvature drift of the finite-action folds and prove a triangular-array Gaussian limit in the complementary hard-cutoff regime. All statements include conventional proofs. Exact coefficient checks and reproducible numerical diagnostics accompany the article.

Scope and research status. This is a proved model-specific extension of the repository’s finite-action critical-inversion program, not a claimed solution of unrestricted transseries realization or resurgence. Stable limits for simply generated trees and soft/hard truncation are established subjects; those mechanisms are credited below. Global publication priority for the combined results has not been established. No new Lean formalization is claimed.

Contents

1	The gap, the contribution, and the classical background	3
1.1	What changes when the action family is countable	3
1.2	What is classical, and what is supplied here	3
2	Model, coordinates, and exact coefficients	4
3	A convergent Hahn inverse at the critical boundary	5
3.1	A closed formula for every coefficient	6
4	Coupling transition and the three coefficient regimes	7
4.1	A uniform local coupling profile	8
5	Stable densities and all-order sector asymptotics	9
6	An exact Hahn–Fourier identity and the cancellation sieve	11
7	A sharp joint coupling–action-cutoff profile	12
7.1	Why the profile is a genuine conditional probability	14
8	How finite-action folds approach the fractional boundary	15
8.1	The two limits really differ	18
9	Reproducible computation and numerical diagnostics	18
9.1	What was checked exactly	18
9.2	The prototype with exponent $\alpha = 3/2$	18
9.3	Reproduction and reliability boundaries	19
10	Interpretation for transseries and a formalization route	20
10.1	Three scales, kept separate	20
10.2	Finite-prefix universality has a precise boundary	20
10.3	What could be formalized first	20
11	Further research questions and concrete targets	21
12	Conclusion	23
A	Source audit and proof dependencies	23
A.1	Repository snapshot and limits of inspection	23
A.2	Dependency structure	24
B	A compact notation ledger	24

1 The gap, the contribution, and the classical background

1.1 What changes when the action family is countable

The repository contains a substantial inversion calculus and recent research packages on support-controlled reversion, analytic moving folds, and regularity of countable exponential feedback. The moving-fold package constructs a local two-sheet atlas for an analytic finite-action equation; its research questions include higher critical multiplicity, changing actions, and the loss of a uniform action gap. The regularity package studies $U(q) = \sum_{j \geq 1} q^j e^{\lambda_j U(q)}$ and separates formal existence, convergence, Gevrey regularity, and realization [5, 6, 7].

Editorial note (ProveIt, 2026-09-29). The finite-action result continued here is Theorem `thm:actions` of the moving-fold article [6]. The model of Section 2, $U = \beta \sum_j a_j q^j e^{jU}$, is the slice $c_j = \beta a_j$, q -exponents and slopes both equal to j , of the regularity article’s further-research question “Amplitudes, general actions, and multiple scales” (`Exponential_Feedback_Regularity_Classification/article.tex`, Section `sec:future`), which asks for $U = \sum_j c_j q^{a_j} e^{\lambda_j U}$ with general amplitudes. This article does not cite that question. For this slice the solution converges, and the article determines its critical boundary and exponent $1/\alpha$; the question’s analogue of the exact type identity for general (c_j, a_j, λ_j) is not addressed.

Here the slopes are only linear, but the coefficients have a power-law tail. The solution therefore converges near the origin, while its first critical boundary is not analytic. The second moment diverges. The critical exponent becomes $1/\alpha$, rather than $1/2$, and a finite truncation of the action family restores a square root at a moving critical value. Consequently, taking the sector index to infinity and taking the action cutoff to infinity do not commute without a scale condition. This is the specific gap addressed here.

We do not identify a nonanalytic boundary with an analytic critical point of a nonintegral “multiplicity.” Those are different objects. Nor do we claim to settle every higher-multiplicity or parameter-dependent-action question in the moving-fold manuscript. Our results give a precise, self-contained answer for the class defined in Section 2.

1.2 What is classical, and what is supplied here

The substitution $t = qe^U$ turns the problem into $t = q \exp(\beta A(t))$, an instance of the classical simply generated tree schema. Janson’s survey gives critical infinite-variance local limits and stable maximum-size behavior; in particular, Example 18.27 treats a closely related power-law allocation setting [3]. Soft and hard truncation of heavy tails are studied by Chakrabarty and Samorodnitsky [4]. Lagrange inversion, singularity transfer, and saddle methods are standard tools [2]. We do not relabel these general principles as new discoveries.

The contribution developed in this article is an explicit connection between the analytic and sector descriptions of this countable-action model. It consists of the following linked results.

1. A convergent critical Hahn inverse, its closed multi-index formula, its exact ramification, and a geometric truncation estimate (Theorems 3.1 and 3.3).
2. An arbitrary-order stable-density expansion and an exact Hahn–Fourier coefficient identity, including an arithmetic cancellation rule (Theorems 5.2 and 6.1).
3. A two-parameter coupling–cutoff profile and a necessary and sufficient action-budget condition (Theorems 7.2 and 7.3).
4. Explicit finite-fold drift constants and a uniform Gaussian coefficient law when the cutoff lies below the critical budget (Theorems 8.1 and 8.2).

These are intended as a reusable extension within the repository. The proofs do not depend on the correctness of any unreviewed repository manuscript. An independent literature and mathematical review remains necessary before asserting publication novelty.

2 Model, coordinates, and exact coefficients

Fix

$$1 < \alpha < 2, \quad a > 0, \quad A(t) = a \operatorname{Li}_{1+\alpha}(t) + P(t) = \sum_{j \geq 1} a_j t^j, \quad (2.1)$$

where $P(t) = \sum_{j=1}^J p_j t^j$, every a_j is nonnegative, and $a_1 > 0$. The coefficients of P itself need not be nonnegative. Thus an arbitrary finite prefix can be changed while the positive tail is held fixed. Put

$$d_k = a\zeta(1 + \alpha - k) + \sum_{j=1}^J p_j j^k \quad (k \geq 0), \quad G = \Gamma(-\alpha) > 0. \quad (2.2)$$

The quantities $d_0 = A(1)$ and $d_1 = \sum j a_j$ are finite and positive. For $k \geq 2$, the numbers d_k are *analytic expansion coefficients*, not finite moments of the action distribution. In particular, $\sum j^2 a_j = \infty$, even though d_2 is a finite real number.

For $\beta > 0$, define $U_\beta \in q\mathbb{R}[[q]]$ by

$$U_\beta(q) = \beta A(qe^{U_\beta(q)}), \quad u_n(\beta) = [q^n]U_\beta(q). \quad (2.3)$$

Its coefficient at degree n depends only on earlier coefficients and $a_1, \dots, a_n$, so the formal solution is unique. The analytic implicit function theorem at $(q, U) = (0, 0)$ also gives a convergent solution there. All its coefficients are positive, since $a_1 > 0$.

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. The central parameters are

$$\beta_c = d_1^{-1}, \quad \delta_\beta = 1 - \beta d_1, \quad \rho_\beta = e^{-\beta d_0}, \quad B_\beta = \beta a G. \quad (2.4)$$

We also write $\rho_c = \rho_{\beta_c}$ and $B_c = B_{\beta_c}$.

Proposition 2.1 (Exact parametrization and Lagrange coefficient). *With $t = qe^{U_\beta(q)}$ one has*

$$U_\beta = \beta A(t), \quad q = te^{-\beta A(t)}, \quad u_n(\beta) = \frac{1}{n} [t^n] e^{n\beta A(t)} \quad (n \geq 1). \quad (2.5)$$

If N_j are independent Poisson random variables of means $n\beta a_j$ and $K_n = \sum_{j \geq 1} j N_j$, then

$$u_n(\beta) = \frac{\rho_\beta^{-n}}{n} \mathbb{P}(K_n = n), \quad \mathbb{E}K_n = n\beta d_1. \quad (2.6)$$

Proof. The first two identities follow directly from (2.3). Lagrange inversion for $t = qe^{\beta A(t)}$ gives

$$[q^n] \beta A(t(q)) = \frac{1}{n} [t^{n-1}] \beta A'(t) e^{n\beta A(t)} = \frac{1}{n} [t^n] e^{n\beta A(t)},$$

where the last equality follows by differentiating the exponential. The total Poisson intensity is $n\beta d_0 < \infty$, so only finitely many N_j are nonzero almost surely. The probability generating function of K_n is $\exp(n\beta(A(t) - d_0))$. Comparing its coefficient at t^n proves (2.6); the mean follows from $d_1 < \infty$. $\square$

For an integer $M \geq 1$, let $A_M(t) = \sum_{j \leq M} a_j t^j$ and let $U_{\beta, M}$ and $u_{n, M}(\beta)$ denote the corresponding solution and coefficients. A particularly useful consequence is

$$\frac{u_{n, M}(\beta)}{u_n(\beta)} = \mathbb{P}(N_j = 0 \text{ for all } j > M \mid K_n = n). \quad (2.7)$$

Indeed, independence contributes the factor $\exp(-n\beta \sum_{j > M} a_j)$, and the remaining coefficient is the truncated probability in (2.6). The ratio belongs to $[0, 1]$, is nondecreasing in M , and equals 1 for $M \geq n$. This identity will turn an analytic action-cutoff question into a conditional large-jump question, without approximation.

3 A convergent Hahn inverse at the critical boundary

The polylogarithm expansion [1, Eq. 25.12.12], on the branch $|\arg x| < \pi$ and $|x| < 2\pi$, gives

$$A(e^{-x}) = d_0 - d_1 x + a G x^\alpha + \sum_{k \geq 2} \frac{d_k (-x)^k}{k!}. \quad (3.1)$$

The series on the right is convergent in that disk. Introduce

$$x = -\log t, \quad \varepsilon = \log(\rho_\beta/q), \quad H_\beta(x) = \beta \sum_{k \geq 2} \frac{d_k (-1)^k x^{k-2}}{k!}. \quad (3.2)$$

Equations (2.5) and (3.1) become the exact local normal form

$$\boxed{\varepsilon = \delta_\beta x + B_\beta x^\alpha + x^2 H_\beta(x), \quad U_\beta = \beta d_0 + \varepsilon - x.} \quad (3.3)$$

For positive $q < \rho_c$ at criticality, x and ε are positive. Complex branches below are continuations of this positive branch.

Theorem 3.1 (Convergent critical Hahn chart). *At $\beta = \beta_c$, set*

$$T = (\varepsilon/B_c)^{1/\alpha}, \quad R = T^{2-\alpha}.$$

There is a holomorphic function $V(T, R)$ near $(0, 0)$, with $V(0, 0) = 1$, such that the critical inverse is exactly

$$x = TV(T, T^{2-\alpha}), \quad V^\alpha + \frac{R}{B_c} V^2 H_{\beta_c}(TV) = 1. \quad (3.4)$$

Consequently its Hahn support in ε is contained in

$$\left\{ \frac{1 + k + (2 - \alpha)m}{\alpha} : k, m \in \mathbb{N} \right\}. \quad (3.5)$$

The expansion is absolutely convergent on a sufficiently small slit neighborhood, and its support has only finitely many elements below any fixed bound.

More quantitatively, suppose V is bounded by K_0 on a polydisk with radii T_0, R_0 . If its Taylor polynomial of total degree at most N is $V_{\leq N}$, and

$$\theta = \max\{|T|/T_0, |T|^{2-\alpha}/R_0\} < 1,$$

then

$$|x - TV_{\leq N}(T, T^{2-\alpha})| \leq |T| K_0 \frac{(N+2)\theta^{N+1}}{(1-\theta)^2}. \quad (3.6)$$

Proof. Substitute $x = TV$ into (3.3) with $\delta_\beta = 0$ and divide by $B_c T^\alpha$. This gives (3.4). Its derivative with respect to V at $(T, R, V) = (0, 0, 1)$ is $\alpha \neq 0$. The analytic implicit function theorem gives the claimed V and its Taylor series. Multiplication by T and then substitution $R = T^{2-\alpha}$ give (3.5). Both grid generators are positive. Thus a bound $1 + k + (2 - \alpha)m \leq C$ bounds k and m separately, proving local finiteness even when the additive group they generate is dense.

Cauchy's estimate bounds the Taylor coefficient of $T^k R^m$ by $K_0 T_0^{-k} R_0^{-m}$. At total degree ℓ there are $\ell + 1$ such monomials. The omitted sum is therefore at most

$$|T|K_0 \sum_{\ell > N} (\ell + 1)\theta^\ell \leq |T|K_0(N + 2)\theta^{N+1}(1 - \theta)^{-2}.$$

This also proves absolute convergence after the substitution. $\square$

Remark 3.2 (Obtaining actual majorant radii). The radii in the theorem need not remain purely existential. Choose $0 < r_0 < 2\pi$ and a bound $H_0 \geq \max(1, \sup_{|x| \leq r_0} |H_{\beta_c}(x)|)$. On $|v - 1| = 1/4$, put $m_\alpha = \min |v^\alpha - 1| > 0$, using the branch near 1. One may take

$$T_0 = r_0/2, \quad R_0 = \frac{B_c m_\alpha}{4(5/4)^2 H_0}, \quad K_0 = 5/4.$$

On that v -circle the perturbation has modulus at most $m_\alpha/4$. Rouché's theorem gives exactly one zero, hence a simple zero depending holomorphically on (T, R) , inside the circle. Slightly enlarging the parameter radii preserves the strict inequality, so Cauchy estimates at the displayed radii are legitimate. Evaluating H_0 and m_α with outward rounding would produce numerical certificates; the supplied numerical program does not perform that interval-arithmetic step.

3.1 A closed formula for every coefficient

For a finitely supported multi-index $\mathbf{m} = (m_2, m_3, \dots)$ define

$$m_* = \sum_{k \geq 2} m_k, \quad K = \sum_{k \geq 2} k m_k, \quad \Lambda = 1 + K - \alpha m_*, \quad h_k = \frac{\beta_c d_k (-1)^k}{k! B_c}. \quad (3.7)$$

In particular $\Lambda \geq 1 + (2 - \alpha)m_* > 0$.

Theorem 3.3 (Exact multi-index reversion). *The chart in Theorem 3.1 has the absolutely convergent expansion*

$$x = \sum_{\mathbf{m}} C_{\mathbf{m}} \left(\prod_{k \geq 2} h_k^{m_k} \right) T^\Lambda, \quad C_{\mathbf{m}} = \frac{(-1)^{m_*}}{\alpha \prod_{k \geq 2} m_k!} \frac{\Gamma((K + 1)/\alpha)}{\Gamma(1 + \Lambda/\alpha)}. \quad (3.8)$$

The zero multi-index contributes T . Terms with equal exponent are to be grouped; every such group is finite.

Proof. Write $s = x/T$ and $y = s^\alpha$. The equation for y is

$$y = 1 - \sum_{k \geq 2} z_k y^{k/\alpha}, \quad z_k = h_k T^{k-\alpha}.$$

First retain only finitely many z_k and introduce an auxiliary variable w multiplying their sum. Lagrange inversion about $y = 1$, applied to $y^{1/\alpha}$, says that the coefficient of a monomial of total degree $m_* > 0$ is

$$\frac{(-1)^{m_*}}{\alpha \prod m_k!} \left. \frac{d^{m_*-1}}{dy^{m_*-1}} y^{(K+1)/\alpha-1} \right|_{y=1}.$$

The derivative is $\Gamma((K+1)/\alpha)/\Gamma((K+1)/\alpha - m_* + 1)$, which gives (3.8). The formula for $m_* = 0$ also yields 1 by $\Gamma(z+1) = z\Gamma(z)$.

A bound on Λ bounds m_* and every index k appearing in $\mathbf{m}$, so passing from finitely many z_k to all of them is coefficientwise finite. To justify absolute rather than just formal convergence, replace each h_k by $-|h_k|$. The same implicit-function argument applies to the analytic majorant $\sum |h_k|T^{k-2}$. The gamma factors in (3.8) are positive, so every coefficient of the resulting majorant expansion is $|C_{\mathbf{m}}| \prod |h_k|^{m_k}$. Its convergence proves absolute convergence of the original expansion and validates grouping. $\square$

For example, the terms supported only on h_2 start as

$$x = T - \frac{h_2}{\alpha} T^{3-\alpha} + \frac{5-\alpha}{2\alpha^2} h_2^2 T^{5-2\alpha} + \dots, \quad (3.9)$$

while the term linear in h_3 is $-(h_3/\alpha)T^{4-\alpha}$. These displayed families must be sorted together; their relative order can depend on α .

Corollary 3.4 (Exact ramification and exponential action grid). *If $\alpha = r/s$ in lowest terms, the critical inverse becomes holomorphic after $\varepsilon = z^r$, and r is the smallest possible positive integer ramification. If α is irrational, no finite Puiseux cover makes the inverse holomorphic.*

If $\varepsilon = e^{-\omega X}$ with $\omega > 0$ and $X \rightarrow +\infty$, (3.8) is an absolutely convergent exponential Hahn transseries. A sufficient positive action grid for x has initial action ω/α and increments ω/α and $\omega(2-\alpha)/\alpha$. The analytic term ε in U is included separately.

Proof. For $\alpha = r/s$, T is a constant times z^s and $T^{2-\alpha}$ is a constant times z^{2s-r} . Both exponents are positive integers. The leading term of x is a nonzero multiple of $\varepsilon^{s/r}$, so any finite ramification must be divisible by r . If α is irrational, the leading exponent $1/\alpha$ cannot be made integral by multiplying it by any positive integer. The exponential statement follows by direct substitution in the convergent grid. $\square$

4 Coupling transition and the three coefficient regimes

The fractional boundary is the transition between an inherited power-law singularity and an interior analytic fold.

Theorem 4.1 (Phase diagram). *For fixed $\beta > 0$, the radius and leading coefficients are as follows. If $0 < \beta < \beta_c$, the radius is ρ_β and*

$$u_n(\beta) \sim \beta a \delta_\beta^{-1-\alpha} \rho_\beta^{-n} n^{-1-\alpha}. \quad (4.1)$$

At $\beta = \beta_c$, the radius is ρ_c and

$$u_n(\beta_c) \sim \rho_c^{-n} B_c^{-1/\alpha} \frac{-1}{\Gamma(-1/\alpha)} n^{-1-1/\alpha}. \quad (4.2)$$

If $\beta > \beta_c$, there is a unique $\tau \in (0, 1)$ with $\beta \tau A'(\tau) = 1$. Writing

$$r_\beta = \tau e^{-\beta A(\tau)}, \quad v_\beta = \beta (t \partial_t)^2 A(t)|_{t=\tau} > 0, \quad (4.3)$$

one has radius r_β and

$$u_n(\beta) \sim \frac{r_\beta^{-n}}{\sqrt{2\pi v_\beta} n^{3/2}}. \quad (4.4)$$

Proof. The function $\beta t A'(t)$ is strictly increasing for real $0 < t < 1$, starts at 0, and tends to βd_1 . Thus $q(t) = t e^{-\beta A(t)}$ is increasing up to $t = 1$ when $\beta \leq \beta_c$, and has its unique first maximum at $t = \tau < 1$ when $\beta > \beta_c$.

For $\delta_\beta > 0$, inversion of (3.3) gives

$$x = \varepsilon / \delta_\beta - B_\beta \delta_\beta^{-1-\alpha} \varepsilon^\alpha + o(\varepsilon^\alpha).$$

The first nonanalytic term of U_β is therefore $B_\beta \delta_\beta^{-1-\alpha} \varepsilon^\alpha$. Singularity transfer gives its coefficient divided by $\Gamma(-\alpha)$, which proves (4.1). At criticality the leading nonanalytic term is $-B_c^{-1/\alpha} \varepsilon^{1/\alpha}$ by Theorem 3.1, proving (4.2).

At the interior critical point put $t = \tau e^{-x}$ and $\varepsilon = \log(r_\beta/q)$. Taylor expansion gives $\varepsilon = v_\beta x^2/2 + O(x^3)$ and $U_\beta = \beta A(\tau) + \varepsilon - x$. The singular term is $-\sqrt{2/v_\beta} \varepsilon^{1/2}$. Since $\Gamma(-1/2) = -2\sqrt{\pi}$, this yields (4.4).

For completeness, these local expansions describe the actual unique dominant singularity. Pringsheim's theorem excludes a smaller convergence radius, since the positive real parametric branch is analytic strictly before the stated boundary. The nonanalytic local term excludes a larger radius. The coefficients of $t(q)$ are positive, with its first two coefficients nonzero. Monotone convergence along the positive radius gives a finite boundary value, either 1 or τ . At every other point on the same circle, strictness in the triangle inequality gives $|t(q)|$ smaller than that positive value. Consequently $|\beta t A'(t)| < 1$ there, and the implicit function theorem continues the solution across that point. Near the positive point the local power-fractional or square-root inverse continues in a slit sector, by (3.3) or the analytic fold expansion. Compactness of the rest of the circle gives a common indented continuation domain. The usual singularity-transfer contour is therefore valid [2, Ch. VI]; no unverified competing singularity is discarded. $\square$

4.1 A uniform local coupling profile

The same normal form gives a useful transition in the *argument* coordinate, separate from the large-sector transition below.

Proposition 4.2 (Boundary-layer inverse and emerging fold). *Suppose $\varepsilon \downarrow 0$ and*

$$\delta_\beta = \eta B_c^{1/\alpha} \varepsilon^{(\alpha-1)/\alpha}, \quad \eta \text{ in a fixed compact subset of } \mathbb{R}.$$

There is a unique positive solution near the critical branch, and

$$x = (\varepsilon/B_c)^{1/\alpha} (v(\eta) + O(\varepsilon^\kappa)), \quad v(\eta)^\alpha + \eta v(\eta) = 1, \quad \kappa = \frac{\min(\alpha-1, 2-\alpha)}{\alpha}. \quad (4.5)$$

The error is uniform on that compact set. As $\delta_\beta \uparrow 0$ through negative values, the emerging interior fold satisfies

$$x_f \sim \left(\frac{-\delta_\beta}{\alpha B_c} \right)^{1/(\alpha-1)}, \quad (4.6)$$

$$\log(r_\beta/\rho_\beta) \sim (\alpha-1)\alpha^{-\alpha/(\alpha-1)} B_c^{-1/(\alpha-1)} (-\delta_\beta)^{\alpha/(\alpha-1)}. \quad (4.7)$$

Proof. Divide (3.3) by ε , putting $x = (\varepsilon/B_c)^{1/\alpha} v$. Since $B_\beta/B_c = \beta/\beta_c = 1 - \delta_\beta$, the result is

$$1 = \eta v + v^\alpha + O(\varepsilon^{(\alpha-1)/\alpha}) v^\alpha + O(\varepsilon^{(2-\alpha)/\alpha}) v^2.$$

The positive root of $v^\alpha + \eta v = 1$ is unique: for $\eta < 0$ the function first decreases below zero and then increases through 1 only once; for $\eta \geq 0$ it increases throughout. At that root its derivative is $1/v + (\alpha - 1)v^{\alpha-1} > 0$. On compact η -sets the root and this derivative are uniformly controlled, proving (4.5).

At the interior fold the derivative of the right side of (3.3) vanishes. Its first two terms are $\delta_\beta + \alpha B_\beta x^{\alpha-1}$; the analytic remainder is $O(x) = o(x^{\alpha-1})$. This proves (4.6). Substitution back into the normal form gives $\varepsilon_f \sim -(\alpha - 1)B_c x_f^\alpha$. Since $\varepsilon_f = \log(\rho_\beta/r_\beta)$, (4.7) follows. $\square$

5 Stable densities and all-order sector asymptotics

We use the principal power $(-iu)^\alpha$ for real u . Because $\cos(\pi\alpha/2) < 0$, the integral

$$g_\alpha(z) = \frac{1}{2\pi} \int_{\mathbb{R}} \exp((-iu)^\alpha - iuz) du \quad (5.1)$$

converges absolutely, together with every derivative in z . This is the density of a centered, spectrally positive stable law in a normalization chosen to match the feedback equation.

Lemma 5.1 (Normalization, derivatives, and positivity). *The exponent in (5.1) has the Lévy representation*

$$(-iu)^\alpha = \frac{1}{G} \int_0^\infty (e^{iux} - 1 - iux)x^{-1-\alpha} dx. \quad (5.2)$$

The function g_α is a smooth probability density, positive for every real argument. At the origin,

$$g_\alpha^{(k)}(0) = \frac{\Gamma((k+1)/\alpha)}{\pi\alpha} \sin \frac{\pi(k+1)}{\alpha}, \quad g_\alpha(0) = -\frac{1}{\Gamma(-1/\alpha)} > 0. \quad (5.3)$$

Proof. For $\Re s > 0$, two integrations by parts give

$$\int_0^\infty (e^{-sx} - 1 + sx)x^{-1-\alpha} dx = \frac{\Gamma(2-\alpha)}{\alpha(\alpha-1)} s^\alpha = \Gamma(-\alpha)s^\alpha.$$

Pass to $s = -iu$ by continuity; integrability near zero follows from the quadratic cancellation, and at infinity from $\alpha > 1$. This proves (5.2). The centered compensated Poisson construction on $(0, 1]$, followed by the finite-intensity jumps on $(1, \infty)$, realizes this characteristic function. Fourier integrability then gives the smooth density in (5.1).

For the derivative formula, restrict the Fourier integral to $u > 0$ and take twice its real part. With $c = -e^{-i\pi\alpha/2}$, which has positive real part, substitute $w = u^\alpha$ to obtain

$$g_\alpha^{(k)}(0) = \frac{\Gamma((k+1)/\alpha)}{\pi\alpha} \Re((-i)^k c^{-(k+1)/\alpha}).$$

The real part is $\sin(\pi(k+1)/\alpha)$, giving the first identity. Gamma reflection yields the second.

Here is also a direct positivity argument useful later. For any $L > 0$ let X_L have the compensated Lévy measure $G^{-1}x^{-1-\alpha}\mathbf{1}_{(0,L]}(x) dx$. Its density exists by the Fourier estimates in Lemma 7.1 below. Split it at $\epsilon < L$:

$$X_L = X_\epsilon + S_{\epsilon,L} - \frac{1}{G} \int_\epsilon^L x^{-\alpha} dx,$$

where $S_{\epsilon,L}$ is an independent compound Poisson sum. The variance of X_ϵ tends to zero, while the displayed compensation tends to positive infinity as $\epsilon \downarrow 0$. Conditional on a sufficiently large finite number of jumps, $S_{\epsilon,L}$ has a density positive throughout the interval between that

number times ϵ and that number times L . These intervals overlap for large numbers of jumps. For any prescribed real z , choose ϵ small enough that z plus the compensation lies strictly inside such an interval. Convolution with the positive-probability event that X_ϵ is close to zero then gives a strictly positive density at z . Thus $g_{\alpha,L}(z) > 0$. Adding the jumps above L to X_L , and using the positive probability of no such jumps, proves $g_\alpha(z) > 0$. This argument does not assume positivity merely from Fourier integrability. $\square$

For a coupling β , which may now depend on n , put

$$b_n = (n\beta aG)^{1/\alpha}, \quad \lambda_n = \frac{n(1 - \beta d_1)}{b_n}, \quad c_{k,n} = \frac{n\beta d_k}{k!b_n^k} \quad (k \geq 2). \quad (5.4)$$

The natural critical coupling window is $\lambda_n = O(1)$, equivalently $\beta - \beta_c = O(n^{-(\alpha-1)/\alpha})$.

Theorem 5.2 (Uniform all-order stable expansion). *Fix a compact interval $I \subset (0, \infty)$ for β and $R > 0$. For a finite multi-index $\mathbf{m}$ as in (3.7), put*

$$w(\mathbf{m}) = \frac{K - \alpha m_*}{\alpha} = \sum_{k \geq 2} \frac{k - \alpha}{\alpha} m_k. \quad (5.5)$$

Uniformly for $\beta \in I$, as $n \rightarrow \infty$,

$$\boxed{nb_n \rho_\beta^n u_n(\beta) = \sum_{w(\mathbf{m}) < R} \frac{\prod_{k \geq 2} c_{k,n}^{m_k}}{\prod_{k \geq 2} m_k!} (-1)^K g_\alpha^{(K)}(\lambda_n) + O_{I,R}(n^{-R})}. \quad (5.6)$$

The sum is finite. The error is an absolute, additive error in the displayed normalization; it is not a relative estimate uniformly far from the critical window. In particular, if $\lambda_n \rightarrow \lambda \in \mathbb{R}$,

$$u_n(\beta) \sim \frac{\rho_\beta^{-n}}{nb_n} g_\alpha(\lambda). \quad (5.7)$$

Proof. We provide the Fourier estimates explicitly. By Proposition 2.1, the left side of (5.6) is $b_n \mathbb{P}(K_n = n)$. Fourier inversion on the lattice gives

$$b_n \mathbb{P}(K_n = n) = \frac{1}{2\pi} \int_{-\pi b_n}^{\pi b_n} \chi_n(u) e^{-i\lambda_n u} du, \quad (5.8)$$

$$\chi_n(u) = \exp \left(n\beta \left[A(e^{iu/b_n}) - d_0 - id_1 u/b_n \right] \right).$$

For $|u|/b_n < 2\pi$, (3.1) implies the exact local expansion

$$\chi_n(u) = \exp \left((-iu)^\alpha + \sum_{k \geq 2} c_{k,n} (iu)^k \right). \quad (5.9)$$

The needed bounds on the rest of the Fourier contour follow from the positive tail, not from a formal Taylor series. There exist $c > 0$ and $\theta_0 > 0$, independent of $\beta \in I$, such that

$$\sum_{j \geq 1} a_j (1 - \cos(j\theta)) \geq \begin{cases} c|\theta|^\alpha, & 0 < |\theta| \leq \theta_0, \\ c, & \theta_0 \leq |\theta| \leq \pi. \end{cases} \quad (5.10)$$

For the first bound, sum only over a block of integers with $1 \leq j|\theta| \leq 2$; they are in the unchanged tail when θ_0 is small. Each cosine defect is bounded below, and the sum of $j^{-1-\alpha}$ over that block is comparable to $|\theta|^\alpha$. For the second bound, the single coefficient $a_1 > 0$ suffices

on the compact interval shown. Thus $|\chi_n(u)| \leq e^{-c'|u|^\alpha}$ for $|u| \leq \theta_0 b_n$, and $|\chi_n(u)| \leq e^{-c'n}$ beyond that interval.

Choose a central cutoff $L_n = (C \log n)^{1/\alpha}$, with C large enough in terms of R . The contribution from $|u| > L_n$ to (5.8) is $O(n^{-R})$, and so are the tails of all finitely many stable-density derivative integrals used below. On $|u| \leq L_n$, the analytic perturbation in (5.9) is $O(n^{-(2-\alpha)/\alpha}|u|^2)$ and tends to zero uniformly. The analytic series can be cut after an index k_0 with $(k_0 - \alpha)/\alpha > R$. Cauchy bounds for its coefficients show that the omitted part is bounded by a constant times $n^{-R}(1 + |u|^{k_0+1})$ on this central interval, after increasing k_0 if necessary.

Expand the exponential of the remaining finite polynomial. Each multi-index contributes exactly $\prod c_{k,n}^{m_k}(iu)^K / \prod m_k!$, with size $O(n^{-w(m)}|u|^K)$. Because every basic weight $(k - \alpha)/\alpha$ is positive, only finitely many have weight below R . Taylor's remainder and the finite analytic coefficient bounds leave an error of the form

$$C_R n^{-R} (1 + |u|^{D_R}) e^{-c''|u|^\alpha}$$

after multiplication by the stable factor, for a finite integer D_R . This bound is integrable with a constant independent of n and β . It proves the asserted remainder without introducing unwanted powers of $\log n$.

Finally,

$$\frac{1}{2\pi} \int_{\mathbb{R}} (iu)^K e^{(-iu)^\alpha - i\lambda_n u} du = (-1)^K g_\alpha^{(K)}(\lambda_n).$$

In all estimates the oscillatory factor $e^{-i\lambda_n u}$ has modulus one, so they are uniform even when λ_n is unbounded. This proves the additive statement. For bounded λ_n , positivity and continuity of g_α give a positive lower bound on compact intervals, which justifies the relative conclusion (5.7). $\square$

Remark 5.3 (Why the qualification on the error matters). For fixed subcritical or supercritical β , $|\lambda_n|$ diverges. The leading density can then be smaller than the additive remainder in (5.6). One cannot obtain a uniform relative theorem in those regimes merely by dividing that equation by $g_\alpha(\lambda_n)$. The fixed-coupling results in Theorem 4.1 have a separate singularity proof. A quantitative matching theorem on an expanding λ -window is left as a research problem.

6 An exact Hahn–Fourier identity and the cancellation sieve

The local inverse and the sector expansion are not two unrelated approximations. Their coefficients match term by term.

Theorem 6.1 (Coefficient duality). *At criticality, the multi-index $\mathbf{m}$ contributes to x the term*

$$C_{\mathbf{m}} \left(\prod h_k^{m_k} \right) B_c^{-\Lambda/\alpha} \varepsilon^{\Lambda/\alpha}.$$

Its nonanalytic contribution to u_n has multiplier

$$-\frac{C_{\mathbf{m}} \prod h_k^{m_k} B_c^{-\Lambda/\alpha}}{\Gamma(-\Lambda/\alpha)} \quad (6.1)$$

in front of $\rho_c^{-n} n^{-1-\Lambda/\alpha}$. This multiplier equals

$$\frac{(-1)^K g_\alpha^{(K)}(0)}{\prod m_k!} \beta_c^{m_*} B_c^{-(K+1)/\alpha} \prod_{k \geq 2} \left(\frac{d_k}{k!} \right)^{m_k}, \quad (6.2)$$

which is exactly the corresponding term of Theorem 5.2 after division by nb_n .

The universal gamma factor vanishes precisely when $(K+1)/\alpha$ is an integer. If $\alpha = r/s$ in lowest terms this means $r \mid K+1$. Equivalently, that Hahn term is an integer power of ε and is locally analytic. For irrational α this particular cancellation never occurs. Other cancellations caused by zero d_k or by grouping equal exponents are not excluded.

Proof. In the logarithmic coordinate $q = \rho_c e^{-\varepsilon}$, Cauchy's coefficient differential is exactly $q^{-n} dq/q = -\rho_c^{-n} e^{n\varepsilon} d\varepsilon$. The local Hankel integral for a power ε^z therefore has factor $n^{-z-1}/\Gamma(-z)$. There is no additional expansion of a Jacobian. This logarithmic transfer can also be checked against the polylogarithm identity (3.1), whose coefficients are exact powers of n . The minus sign in (6.1) comes from $U = \beta_c d_0 + \varepsilon - x$.

For the algebraic identity, write $z = \Lambda/\alpha = (K+1)/\alpha - m_*$. Gamma reflection gives

$$\frac{1}{\Gamma(1+z)\Gamma(-z)} = -\frac{\sin \pi z}{\pi}, \quad \sin \pi z = (-1)^{m_*} \sin \frac{\pi(K+1)}{\alpha}.$$

Insert (3.8) and $h_k = \beta_c d_k (-1)^k / (k! B_c)$ into (6.1). The resulting sign is $(-1)^K$, the remaining gamma-sine factor is $g_\alpha^{(K)}(0)$ by (5.3), and the power of B_c is $-m_* - \Lambda/\alpha = -(K+1)/\alpha$. This is (6.2).

Since $\Lambda > 0$, the reciprocal gamma factor in (6.1) vanishes exactly when Λ/α is a positive integer. This is equivalent to $(K+1)/\alpha \in \mathbb{N}$. For $\alpha = r/s$ the condition is $r \mid K+1$. The leading monomial coefficients are understood with continuous reciprocal gamma at its zeros, so the identity includes the vanishing cases. $\square$

Example 6.2 (The first correction disappears at $\alpha = 3/2$). For $\alpha = 3/2$, $g_\alpha''(0) = 0$, since $(2+1)/\alpha = 2$. Consequently the nominal relative $n^{-1/3}$ correction $c_{2,n} g_\alpha''(0)$ is absent at critical coupling, regardless of the finite prefix. If $d_2 \neq 0$, the first possible nonzero correction is

$$nb_n \rho_c^n u_n(\beta_c) = g_{3/2}(0) + \frac{c_{2,n}^2}{2} g_{3/2}^{(4)}(0) + O(n^{-1}). \quad (6.3)$$

It has order $n^{-2/3}$. The corresponding first Hahn correction is $-(h_2/\alpha)T^{3-\alpha} = -(h_2/\alpha)T^{3/2}$, an integer power of ε ; hence it contributes no singular large-sector term. This explains the cancellation structurally, rather than discovering it only by a vanishing numerical constant.

Remark 6.3 (Two different kinds of convergence). The critical Hahn expansion in ε is convergent. The expansion in decreasing powers of n in Theorem 5.2 is asserted to all finite orders, not to converge when infinitely many orders are summed. The derivative factors $g_\alpha^{(K)}$ can grow rapidly with K . The identity between their individual coefficients does not identify the two notions of convergence and does not imply Borel summability of the large-sector expansion.

7 A sharp joint coupling-action-cutoff profile

For $L > 0$, define

$$\Psi_{\alpha,L}(u) = \frac{1}{G} \int_0^L (e^{iux} - 1 - iux) x^{-1-\alpha} dx, \quad (7.1)$$

$$g_{\alpha,L}(z) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{\Psi_{\alpha,L}(u) - iuz} du, \quad (7.2)$$

$$\nu_L = \frac{1}{\alpha G L^\alpha}, \quad h_L = \frac{L^{1-\alpha}}{(\alpha-1)G}. \quad (7.3)$$

Here ν_L is the intensity of omitted jumps above L , and h_L is their mean in stable-scale units. Both constants are necessary: deleting large jumps changes the mass and the centering.

Lemma 7.1 (Truncated density estimates). *The function $g_{\alpha,L}$ is a smooth, strictly positive probability density. For L in a compact subinterval of $(0, \infty)$, its defining Fourier integral and all fixed derivatives admit common integrable majorants. As $L \rightarrow \infty$, $g_{\alpha,L} \rightarrow g_\alpha$ locally uniformly. For $0 < L \leq 1$ there is a constant depending only on α such that*

$$\sup_z g_{\alpha,L}(z) \leq C_\alpha L^{-(2-\alpha)/2}. \quad (7.4)$$

Proof. The compensated Poisson construction and the positivity argument in Lemma 5.1 apply with the Lévy measure restricted to $(0, L]$. To justify the density used there, observe that

$$-\Re \Psi_{\alpha,L}(u) = \frac{1}{G} \int_0^L (1 - \cos ux) x^{-1-\alpha} dx \geq c_\alpha \begin{cases} u^2 L^{2-\alpha}, & |u|L \leq 1, \\ |u|^\alpha, & |u|L \geq 1. \end{cases}$$

The first inequality uses $1 - \cos v \geq cv^2$ on $|v| \leq 1$. For the second, restrict to $0 < x < 1/|u|$ and use the first estimate there. These bounds prove integrability with any polynomial factor, hence smoothness and the uniform majorants. They also give

$$\int_{\mathbb{R}} |e^{\Psi_{\alpha,L}(u)}| du \leq CL^{-(2-\alpha)/2} + C,$$

which proves (7.4). As $L \rightarrow \infty$, the exponent converges to (5.2); for $L \geq 1$ the same lower bounds give a common integrable majorant. Dominated convergence proves the local uniform convergence of the densities. $\square$

Theorem 7.2 (Joint critical cutoff law). *Let $\beta = \beta_n$ stay in a compact subinterval of $(0, \infty)$, and use b_n, λ_n from (5.4). Suppose*

$$\lambda_n \longrightarrow \lambda \in \mathbb{R}, \quad \frac{M_n}{b_n} \longrightarrow L \in (0, \infty).$$

Then

$$\boxed{\frac{u_{n,M_n}(\beta_n)}{u_n(\beta_n)} \longrightarrow \mathcal{R}_\alpha(L, \lambda) := e^{-\nu_L} \frac{g_{\alpha,L}(\lambda + h_L)}{g_\alpha(\lambda)}}. \quad (7.5)$$

The convergence is uniform for M_n/b_n and λ_n in fixed compact subintervals of $(0, \infty)$ and $\mathbb{R}$, respectively, in the equivalent form with $\mathcal{R}_\alpha(M_n/b_n, \lambda_n)$ on the right and an $o(1)$ error. The profile is independent of a and of the finite prefix P after these quantities have been incorporated into β_c and b_n .

Proof. Let $K_{n,M} = \sum_{j \leq M} j N_j$ and $\mu_{n,M} = n\beta \sum_{j \leq M} j a_j$. The exact conditional identity can be written

$$\frac{u_{n,M}(\beta)}{u_n(\beta)} = \exp \left(-n\beta \sum_{j > M} a_j \right) \frac{\mathbb{P}(K_{n,M} = n)}{\mathbb{P}(K_n = n)}. \quad (7.6)$$

Because $M \sim Lb_n \rightarrow \infty$, the omitted weights are in the unchanged tail. Integral comparison gives, uniformly on compact L -intervals,

$$n\beta \sum_{j > M} a_j \longrightarrow \nu_L, \quad (7.7)$$

$$\frac{n\beta \sum_{j > M} j a_j}{b_n} \longrightarrow h_L. \quad (7.8)$$

The target centered lattice coordinate is consequently $(n - \mu_{n,M})/b_n = \lambda_n + h_L + o(1)$.

The centered characteristic exponent of $K_{n,M}/b_n$ is

$$n\beta \sum_{j \leq M} a_j (e^{iuj/b_n} - 1 - iuj/b_n).$$

Riemann summation, using $n\beta a/b_n^\alpha = 1/G$, shows its convergence to $\Psi_{\alpha,L}(u)$, locally uniformly in u and L . The integrand at the lower endpoint is integrable after its quadratic cancellation. Explicitly, the contribution of $j \leq \epsilon b_n$ is bounded by $Cu^2\epsilon^{2-\alpha}$, uniformly in n ; on the remainder ordinary Riemann convergence applies. The finitely changed prefix contributes only $O(n/b_n^2) = o(1)$ to the centered exponent.

To upgrade weak convergence to the required lattice local limit, one must control the Fourier tails. For M sufficiently large, unchanged tail weights give

$$\sum_{j \leq M} a_j (1 - \cos j\theta) \geq c \begin{cases} \theta^2 M^{2-\alpha}, & |\theta|M \leq 1, \\ |\theta|^\alpha, & 1/M \leq |\theta| \leq \theta_0. \end{cases} \quad (7.9)$$

For the first bound use a terminal block of integers below M and the quadratic cosine estimate. For the second, use integers up to $1/|\theta|$; after making θ_0 small the unchanged tail supplies a fixed fraction of their second-moment sum. The coefficient a_1 again supplies a uniform gap for $\theta_0 \leq |\theta| \leq \pi$. Upon putting $u = b_n\theta$ and restricting M/b_n to a compact positive interval, these inequalities give a common integrable majorant for the rescaled characteristic functions, with an exponentially small contribution from the macroscopic arc. Fourier inversion and dominated convergence therefore yield

$$b_n \mathbb{P}(K_{n,M} = n) = g_{\alpha, M/b_n} \left(\lambda_n + \frac{n\beta \sum_{j > M} j a_j}{b_n} \right) + o(1) \longrightarrow g_{\alpha, L}(\lambda + h_L).$$

All bounds are uniform on the compact parameter sets specified in the theorem. The denominator is $b_n \mathbb{P}(K_n = n) = g_\alpha(\lambda_n) + o(1)$ by Theorem 5.2, and it stays bounded away from zero there. Insert these estimates and (7.7) into (7.6). This proves (7.5) and its uniform version. The disappearance of the finite prefix from the centered Riemann limit proves the universality assertion. $\square$

7.1 Why the profile is a genuine conditional probability

The Lévy measure above L has finite mass ν_L and first moment h_L . Decomposing the stable variable into jumps below and above L gives the nonnegative density identity

$$g_\alpha(z) = e^{-\nu_L} \sum_{m=0}^{\infty} \frac{1}{m!} \int_{(L, \infty)^m} g_{\alpha, L} \left(z + h_L - \sum_{i=1}^m x_i \right) \prod_{i=1}^m \frac{dx_i}{Gx_i^{1+\alpha}}. \quad (7.10)$$

For $m = 0$, the integral is understood to be $g_{\alpha, L}(z + h_L)$. Thus (7.5) is exactly the fraction of the stable density arising from the event of no jump above L , when conditioned on the final centered value. In particular

$$0 < \mathcal{R}_\alpha(L, \lambda) < 1 \quad (L > 0, \lambda \in \mathbb{R}). \quad (7.11)$$

The first inequality follows from density positivity. The second is strict because the $m = 1$ term in (7.10) is positive. Ignoring either $e^{-\nu_L}$ or the shift h_L would produce the wrong profile.

Theorem 7.3 (Necessary and sufficient critical action budget). *Fix $\beta = \beta_c$ and put $b_n = (nB_c)^{1/\alpha}$. For every sequence of positive integer cutoffs M_n ,*

$$\frac{u_{n,M_n}(\beta_c)}{u_n(\beta_c)} \rightarrow 1 \iff \frac{M_n}{b_n} \rightarrow \infty. \quad (7.12)$$

If $M_n/b_n \rightarrow 0$, the coefficient ratio tends to zero. If $M_n/b_n \rightarrow L \in (0, \infty)$, it tends to $\mathcal{R}_\alpha(L, 0) \in (0, 1)$. The function $L \mapsto \mathcal{R}_\alpha(L, \lambda)$ is nondecreasing and satisfies

$$\lim_{L \downarrow 0} \mathcal{R}_\alpha(L, \lambda) = 0, \quad \lim_{L \rightarrow \infty} \mathcal{R}_\alpha(L, \lambda) = 1 \quad (7.13)$$

for each fixed λ .

Proof. Monotonicity follows by applying (2.7) at two cutoffs $\lfloor L_1 b_n \rfloor \leq \lfloor L_2 b_n \rfloor$ and passing to the limit. For $\lambda \neq 0$, choose the unique nearby coupling satisfying $\lambda_n = \lambda$; the defining expression in (5.4) is strictly decreasing in β , so this choice exists for all large n .

As $L \rightarrow \infty$, $h_L, \nu_L \rightarrow 0$ and $g_{\alpha,L} \rightarrow g_\alpha$ locally uniformly by Lemma 7.1. This proves the second limit in (7.13). For $L \downarrow 0$, the same lemma gives

$$\mathcal{R}_\alpha(L, \lambda) \leq \frac{C_\alpha}{g_\alpha(\lambda)} L^{-(2-\alpha)/2} \exp\left(-\frac{1}{\alpha G L^\alpha}\right) \rightarrow 0.$$

If $M_n/b_n \rightarrow \infty$, compare the coefficient ratio below with the ratio at $\lfloor L b_n \rfloor$ for each fixed L , and above with 1. Theorem 7.2, followed by $L \rightarrow \infty$, proves sufficiency. If $M_n/b_n \rightarrow 0$, compare above with the ratio at $\lfloor L b_n \rfloor$ and then let $L \downarrow 0$. Finally, if M_n/b_n does not tend to infinity, a subsequence is bounded above by some finite L . On that subsequence the upper limit of the coefficient ratio is at most $\mathcal{R}_\alpha(L, 0) < 1$ by (7.11). Thus a ratio tending to 1 forces $M_n/b_n \rightarrow \infty$. The comparable-cutoff case is Theorem 7.2 itself. $\square$

Remark 7.4 (What the budget does and does not mean). Every coefficient of degree n is reproduced *exactly* by retaining n actions. The theorem identifies the much smaller scale needed for asymptotic relative accuracy, but requires a factor tending to infinity beyond $n^{1/\alpha}$ for the ratio to tend to 1. A fixed large multiple can give a very small limiting error, but not zero limiting error. This is a theorem about growing sector index, not about truncating a convergent sum at a fixed numerical value of q .

8 How finite-action folds approach the fractional boundary

At critical coupling every finite truncation has finite variance and an ordinary fold. The location and curvature of that fold contain the same scale that appears in the action-budget theorem.

Let $M \geq J$, and let $\tau_M > 1$ be the unique positive solution of

$$\beta_c \sum_{j \leq M} j a_j \tau_M^j = 1. \quad (8.1)$$

It is larger than 1 because the left side at 1 is less than $\beta_c d_1 = 1$. Define

$$r_M = \tau_M e^{-\beta_c A_M(\tau_M)}, \quad v_M = \beta_c \sum_{j \leq M} j^2 a_j \tau_M^j. \quad (8.2)$$

In particular, $r_M > e^{-\beta_c A_M(1)} > \rho_c$.

There is a unique positive number v_α satisfying

$$\int_0^1 (e^{v_\alpha u} - 1)u^{-\alpha} du = \frac{1}{\alpha - 1}. \quad (8.3)$$

Existence and uniqueness follow because the left side is continuous, strictly increasing from 0 to infinity. Introduce

$$J_\alpha(v) = \frac{1}{\alpha} + \frac{v}{\alpha - 1} - \int_0^1 (e^{vu} - 1 - vu)u^{-1-\alpha} du, \quad (8.4)$$

$$D_\alpha = \beta_c a J_\alpha(v_\alpha), \quad V_\alpha = \beta_c a \int_0^1 u^{1-\alpha} e^{v_\alpha u} du. \quad (8.5)$$

The integrals converge at zero. Moreover, $J'_\alpha(v_\alpha) = 0$ and $J''_\alpha(v) < 0$. Since $J'_\alpha(0) = 1/(\alpha - 1) > 0$, one has $J_\alpha(v_\alpha) > J_\alpha(0) = 1/\alpha$, so $D_\alpha > 0$.

Theorem 8.1 (Critical-value and curvature drift). *As $M \rightarrow \infty$,*

$$M \log \tau_M \longrightarrow v_\alpha, \quad \log(r_M/\rho_c) \sim D_\alpha M^{-\alpha}, \quad v_M \sim V_\alpha M^{2-\alpha}. \quad (8.6)$$

The number v_α depends only on α , and the dependence of D_α, V_α on the finite prefix is only through $\beta_c a$.

Proof. Write $h_M = \log \tau_M$. Subtract $\beta_c \sum_{j \geq 1} j a_j = 1$ from (8.1) to get

$$\sum_{j \leq M} j a_j (e^{h_M j} - 1) = \sum_{j > M} j a_j. \quad (8.7)$$

For any fixed $v \geq 0$, multiply the difference between the two sides, with h_M replaced by v/M , by $M^{\alpha-1}$. Tail summation and a Riemann sum give

$$a \left(\int_0^1 (e^{vu} - 1)u^{-\alpha} du - \frac{1}{\alpha - 1} \right).$$

The finite prefix contributes $O(M^{\alpha-2}) = o(1)$. The convergence is uniform for v on compact intervals. Strict monotonicity in h brackets the root between $(v_\alpha - \epsilon)/M$ and $(v_\alpha + \epsilon)/M$ for every fixed $\epsilon > 0$, proving the first limit in (8.6).

An exact rearrangement, important also for stable numerical evaluation, is

$$\log(r_M/\rho_c) = \beta_c \left(h_M \sum_{j > M} j a_j + \sum_{j > M} a_j - \sum_{j \leq M} a_j (e^{h_M j} - 1 - h_M j) \right). \quad (8.8)$$

Indeed, begin with $h_M + \beta_c(d_0 - A_M(e^{h_M}))$ and use $1 = \beta_c d_1$. After multiplying by M^α , the first two terms in the brackets tend to $av_\alpha/(\alpha - 1)$ and a/α . The last term tends to $a \int_0^1 (e^{v_\alpha u} - 1 - v_\alpha u)u^{-1-\alpha} du$. Its finite-prefix contribution is $O(M^{\alpha-2})$, again negligible. This proves the radius drift with constant D_α .

Finally, Riemann summation in (8.2) gives

$$M^{\alpha-2} v_M \longrightarrow \beta_c a \int_0^1 u^{1-\alpha} e^{v_\alpha u} du = V_\alpha.$$

The integrability at zero permits the same small-initial-block argument as in the proof of Theorem 7.2; finite-prefix terms vanish. $\square$

For fixed M , the analytic fold law is

$$u_{n,M}(\beta_c) \sim \frac{r_M^{-n}}{\sqrt{2\pi v_M} n^{3/2}}. \quad (8.9)$$

The next theorem allows M to increase with n , and identifies the range in which that formula remains valid with its exact moving parameters.

Theorem 8.2 (Uniform fold law in the hard-cutoff regime). *If $M = M_n \rightarrow \infty$ and $n/M_n^\alpha \rightarrow \infty$, then*

$$u_{n,M_n}(\beta_c) = \frac{r_{M_n}^{-n}}{\sqrt{2\pi v_{M_n}} n^{3/2}} (1 + o(1)). \quad (8.10)$$

Thus the finite-fold coefficient law is uniform whenever $M_n = o(n^{1/\alpha})$. At the critical scale $M_n \asymp n^{1/\alpha}$, the non-Gaussian profile of Theorem 7.2 is required instead.

Proof. Exponentially tilt the coefficient identity at $t = \tau_M$. Let $\tilde{N}_j$ be independent Poisson variables of means $n\beta_c a_j \tau_M^j$, $j \leq M$, and set $\tilde{K} = \sum_{j \leq M} j \tilde{N}_j$. Then

$$u_{n,M}(\beta_c) = \frac{r_M^{-n}}{n} \mathbb{P}(\tilde{K} = n), \quad \mathbb{E}\tilde{K} = n, \quad \text{Var}(\tilde{K}) = nv_M. \quad (8.11)$$

Put $\sigma_M = \sqrt{nv_M}$ and $H_n = n/M^\alpha$. By Theorem 8.1,

$$\frac{M}{\sigma_M} \asymp H_n^{-1/2} \rightarrow 0, \quad \tau_M^M \rightarrow e^{v_\alpha}.$$

The third absolute cumulant bound is

$$\frac{n\beta_c \sum_{j \leq M} j^3 a_j \tau_M^j}{\sigma_M^3} = O(H_n^{-1/2}) = o(1).$$

Consequently the centered characteristic function, at argument u/σ_M , converges to $e^{-u^2/2}$ on every bounded u -interval.

For the local, rather than only weak, central limit, split the Fourier integral into three regions. When $|\theta| \leq 1/M$, the quadratic cosine bound gives

$$\left| \mathbb{E} e^{i\theta(\tilde{K}-n)} \right| \leq e^{-cnv_M \theta^2}.$$

After $u = \sigma_M \theta$, this supplies a common Gaussian majorant; its range extends to $\sigma_M/M \asymp \sqrt{H_n} \rightarrow \infty$. For $1/M \leq |\theta| \leq \theta_0$, the second estimate in (7.9), and $\tau_M \geq 1$, give $\left| \mathbb{E} e^{i\theta(\tilde{K}-n)} \right| \leq e^{-cn|\theta|^\alpha}$. The normalized integral over this region is bounded by

$$C\sigma_M n^{-1/\alpha} \int_{n^{1/\alpha}/M}^{\infty} e^{-cv^\alpha} dv \leq CH_n^{1/2-1/\alpha} \int_{H_n^{1/\alpha}}^{\infty} e^{-cv^\alpha} dv = o(1).$$

This estimate is valid even when H_n tends to infinity very slowly. On $\theta_0 \leq |\theta| \leq \pi$, the positive coefficient a_1 gives an e^{-cn} bound. Its integral, multiplied by σ_M , also tends to zero, since the hypothesis bounds M polynomially in n . Fourier inversion now yields $\sigma_M \mathbb{P}(\tilde{K} = n) \rightarrow 1/\sqrt{2\pi}$. Inserting this in (8.11) proves (8.10). $\square$

Remark 8.3 (Do not exponentiate an uncontrolled remainder). Theorem 8.2 uses the *exact* value r_M . It does not permit the replacement $r_M^{-n} = \rho_c^{-n} \exp(-nD_\alpha M^{-\alpha})(1 + o(1))$ throughout its range. Theorem 8.1 only controls $\log(r_M/\rho_c)$ relatively; its unquantified error may be amplified by n . That replacement needs a rate strong enough that $n(\log(r_M/\rho_c) - D_\alpha M^{-\alpha}) \rightarrow 0$. The exact rearrangement (8.8) avoids this problem in computation.

8.1 The two limits really differ

For every fixed M , the power of n in (8.9) is $-3/2$, and the exponential factor has radius $r_M > \rho_c$. For the untruncated critical equation, the power is $-1 - 1/\alpha$ and the radius is ρ_c . The drift $\log(r_M/\rho_c) \asymp M^{-\alpha}$ becomes order one after multiplication by n exactly at $M \asymp n^{1/\alpha}$. This agrees with, but is not a replacement for, the full cutoff profile theorem: the latter supplies both the missing amplitude and the conditioned centering shift. The matching is therefore quantitative, not just a comparison of formal critical exponents.

9 Reproducible computation and numerical diagnostics

9.1 What was checked exactly

The accompanying `code/verify.py` performs three independent finite coefficient calculations: direct substitution into the nonlinear equation, Lagrange's coefficient formula, and a sum over integer partitions. For rational weights it compares the results through degree 13, using Python's exact `Fraction` arithmetic. The test weights are β/j^3 with $\beta = 1/3$, and their cutoff at $M = 4$ with $\beta = 2/5$. The exponent 3 is used to keep the arithmetic rational; these tests verify the algebraic identity, which does not depend on the restriction $1 < \alpha < 2$.

Additional exact comparisons check that retaining M actions preserves every coefficient through degree M . A symbolic calculation at $\alpha = 3/2$ verifies four coefficients of the Hahn implicit equation. In total, **80 exact comparisons passed**. This is evidence for the implementation and the displayed finite identities, not a substitute for the asymptotic proofs.

For a single target sector n , the probability in (2.6) can be computed by a positive recurrence. If p_k is the probability generating coefficient at t^k , with the Poisson parameter n held fixed, then

$$p_0 = e^{-n\beta A_M(1)}, \quad p_k = \frac{n\beta}{k} \sum_{j=1}^{\min(k,M)} j a_j p_{k-j}. \quad (9.1)$$

For the infinite family replace $A_M(1)$ by $A(1)$ and the upper summation limit by k . This costs $O(n \min(n, M))$ arithmetic operations and $O(n)$ storage for one target n . The Poisson parameter changes when the target n changes; this complexity should not be mistaken for a simultaneous all-sector algorithm of the same cost.

9.2 The prototype with exponent $\alpha = 3/2$

For $a = 1$, $P = 0$, and $\alpha = 3/2$, the constants are

$$\begin{aligned} \beta_c &= 0.38279338399942656225\dots, & B_c &= 0.90464481009458340321\dots, \\ \rho_c &= 0.59839006994955853557\dots, & g_{3/2}(0) &= 0.24885478260493015222\dots \end{aligned}$$

They were evaluated with 75-digit arithmetic. The coefficient recurrence was evaluated through target degree 4096 using `numpy.longdouble`; on platforms with a narrower exponent range, the code switches to an arbitrary-precision recurrence if its initial probability underflows. Fourier integrals were computed in double precision, with two quadrature resolutions for the cutoff density. None of these floating-point evaluations is an outward-rounded interval certificate.

In Table 1, $p_n = \mathbb{P}(K_n = n)$ and the corrected value is the right side of (6.3) without its remainder. The disappearance of the nominal $n^{-1/3}$ term is visible in the faster approach to the limiting constant.

n	$b_n p_n$	relative error	corrected value
64	0.247067969	-7.180e-03	0.247226460
256	0.248168803	-2.757e-03	0.248208582
1024	0.248588377	-1.071e-03	0.248598338
4096	0.248750521	-4.190e-04	0.248753012

Table 1: Critical normalized coefficients. The relative error is measured against $g_{3/2}(0)$; the correction is the first nonzero term allowed by the cancellation sieve.

Write $\mathcal{R}_\alpha(L) = \mathcal{R}_\alpha(L, 0)$ at critical coupling. Table 2 compares the limiting retained-coefficient fraction with the actual ratio at $n = 4096$ and $M = \lfloor Lb_n \rfloor$. The latter uses the rounded integer cutoff, so M/b_n is slightly less than the displayed target L . At $L = 1$ a substantial fraction is still lost. A larger fixed multiple can be very accurate but, as Theorem 7.3 states, has a strictly positive limiting loss.

L	$\mathcal{R}_{3/2}(L)$	$n = 4096$ ratio
0.5	0.27103266	0.25750561
1.0	0.76476600	0.76091153
2.0	0.98459916	0.98537953
4.0	0.99999962	0.99999988

Table 2: Critical action-cutoff profile and finite-sector diagnostics. The two quadrature resolutions differed by at most 2.34×10^{-12} in the computed truncated densities; this difference is a diagnostic, not a rigorous error bound.

For the finite-fold constants, the calculations give

$$\begin{aligned} v_{3/2} &= 0.85403265659819698978\dots, \\ D_{3/2} &= 0.59948018701656103250\dots, \\ V_{3/2} &= 1.05291088528937168260\dots \end{aligned}$$

Table 3 illustrates the convergence in Theorem 8.1. The complete data additionally contain the scaled curvatures, critical-coupling-window comparisons at $\lambda = -1, 0, 1$, and cutoff ratios at $n = 256, 1024, 4096$.

M	$M \log \tau_M$	$M^{3/2} \log(r_M/\rho_c)$
16	0.949469078	0.627376226
64	0.906661654	0.619377140
256	0.881417572	0.610854709
1024	0.867971108	0.605513272

Table 3: Moving finite-action folds. The two columns converge to $v_{3/2}$ and $D_{3/2}$, respectively.

9.3 Reproduction and reliability boundaries

Run

```
python -m pip install -r requirements.txt
python code/verify.py --max-n 4096
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A third LaTeX pass may be useful if page references move. The package contains the generated CSV tables and `data/verification.json`, so typesetting does not require rerunning the computation. The formulas and proofs are independent of the precision and quadrature choices in the script. The numerical code does not verify global singularity geometry, local-limit theorems, or the infinite support assertions by finite sampling.

10 Interpretation for transseries and a formalization route

10.1 Three scales, kept separate

The model has three related but distinct asymptotic descriptions. First, near $q = 0$ the original $U_\beta(q) = \sum u_n q^n$ is an ordinary convergent series; substituting $q = e^{-X}$ makes it a convergent integer-action exponential transseries. Second, near its critical boundary one uses $\varepsilon = \log(\rho_c/q)$, and x has the fractional convergent Hahn chart of Section 3. Third, as the original sector index n grows, the coefficients u_n have the stable-density asymptotic expansion of Section 5. Theorem 6.1 relates the last two descriptions. It does not identify their variables.

This separation resolves a potential ambiguity in statements about “closure under inversion.” The small- q analytic germ is harmless; the obstruction to a finite Puiseux chart occurs at a nonanalytic boundary of the countable action sum. At rational α a finite ramification exists and its minimal order is known exactly. At irrational α the critical inverse still has a finite positive Hahn grid, although it has no finite-sheeted Puiseux representation. Thus finite-grid support and finite analytic ramification are different properties.

10.2 Finite-prefix universality has a precise boundary

The finitely many altered weights change d_0 , d_1 , and the analytic coefficients d_k . They therefore change the critical coupling, the exponential growth factor, and the subleading sector corrections. They do not change the leading critical exponents, the admissible Hahn grid, the normalized cutoff profile, or the arithmetic form of the cancellation sieve. The profile theorem is not an assertion that all sequences with the same tail are identical; it identifies exactly which quantities survive the normalization.

The positive-tail assumption supplies the Fourier damping. The condition $a_1 > 0$ is a convenient sufficient aperiodicity hypothesis, not a claim that more general lattice spans are impossible. Replacing it by a nontrivial common divisor would require accessible residue classes and corresponding Fourier aliases. Signed or complex weights would remove the conditional probability interpretation altogether.

10.3 What could be formalized first

No new result in this article has been checked in Lean. The repository’s existing formalization of finite algebraic recurrences or residual brackets does not, by itself, establish the analytic statements here. A realistic staged interface is the following.

The first stage is finite algebra: the coefficient recurrence, Lagrange’s identity for a finite truncation, exact agreement through degree M , and the finite multi-index support bounds. The second stage is scalar special-function algebra: the gamma recurrence and reflection identity proving (6.1)–(6.2), with reciprocal gamma used to include zeros cleanly. The third stage packages the analytic implicit-function and Cauchy-majorant argument with explicit inputs H_0, m_α, T_0, R_0 . These stages isolate checkable certificates without pretending that symbolic coefficient equality proves analytic convergence.

The probability stage should then formalize the independent Poisson coefficient identity and the Fourier tail inequalities. Only after those are in place should one assemble uniform local limits and triangular-array limits. Their most useful reusable content is the integrable-majorant interface, including uniform dependence on compact parameter sets. Finally, the global analytic phase diagram can be linked to a formal singularity-transfer theorem. This is a substantial program, not an already completed formal crosswalk.

11 Further research questions and concrete targets

1. Slowly varying action tails. Replace the exact tail by $a_j \sim aj^{-1-\alpha}\ell(j)$ with slowly varying ℓ . Determine the correct implicit stable scale, the analogue of the cutoff profile, and a usable inverse notation when de Bruijn conjugates are needed. The finite positive Hahn grid may cease to be the right representation; specify sufficient hypotheses that preserve it.

Editorial note (ProveIt, 2026-09-29). Partial progress at the endpoint $\alpha = 2$ only: the logarithmic-endpoint package [8] treats $a_j \sim aj^{-3}(\log j)^r$, $r \geq -1$, at leading order (its `thm:weightedchart`, `thm:weightedbudget`). For $1 < \alpha < 2$ it is answered by the later package `Slowly_Varying_Action_Tails_Critical_Transseries/` (batch 50; not reviewed): scale, cutoff profile and budget for every slowly varying ℓ (its stable scale omits the factor $\Gamma(-\alpha)$ of b_n here; with that dictionary its profile at $\ell \equiv 1$ is Theorem 7.2); a counterexample to the Hahn grid for $\ell = 2 + \sin(\log \log j)$; and a convergent Lambert chart and an effective-index expansion for $(\log j)^m$.

2. Quantitative cutoff errors and certified action selection. Prove an explicit rate in Theorem 7.2, uniform in a chosen rectangle of (L, λ) , and enclose its density integrals by interval arithmetic. The practical target is an algorithm which, given n and a relative tolerance, returns a certified cutoff M . The current profile is exact as a limit, while the supplied quadrature is not certified.

Editorial note (ProveIt, 2026-09-29). Partial progress at the endpoint $\alpha = 2$ only. The logarithmic-endpoint package [8] proves an asymptotic error rate for its refined cutoff law (its `thm:refined`) without explicit constants, and the marginal package [9] proves an explicit finite- n lower bound on the retained fraction (its `thm:finite-certificate`), whose recorded evaluations use no directed rounding. Neither supplies the certified cutoff algorithm asked for here; both list it as open. For $1 < \alpha < 2$ the question is untouched.

3. An expanding coupling window. Find the largest growing range of λ_n on which the all-order expansion is relatively accurate, and match it to the fixed-subcritical power-law regime and the supercritical saddle regime. This requires large-argument estimates for stable-density derivatives and control of how many correction terms are useful; the additive uniform theorem does not already provide it.

4. Endpoint exponents and logarithmic crossover. Study joint limits $\alpha \downarrow 1$ or $\alpha \uparrow 2$ with n . At the upper endpoint the diverging second-moment normalization competes with Gaussian behavior; at the lower endpoint the mean itself is near failure. Derive logarithmic critical charts and cutoff laws rather than substituting an endpoint into formulas containing $\Gamma(-\alpha)$.

Editorial note (ProveIt, 2026-09-29). Answered at the upper endpoint by later packages of this series, which are unrefereed research drafts; the lower endpoint $\alpha \downarrow 1$ is treated by none of them. *Fixed endpoint* $\alpha = 2$, tail aj^{-3} outside a finite prefix: the logarithmic-endpoint

package [8] and the marginal package [9], written independently of each other from a library copy of this article, prove the same core. The pole of $\Gamma(-\alpha)$ cancels the divergent quadratic coefficient and leaves $x^2 \log(1/x)$; the critical inverse is a convergent chart over a Lambert- W_{-1} core; the sector coefficients have an all-order Gaussian–logarithmic expansion; the retained fraction tends to $e^{-1/(2s^2)}$ with the same first correction. Their constants agree after a change of logarithmic normalization ($\ell_n = \log b_n + h$ in the first, $H_n = \log B_n$ in the second), and both agree with the formal $\alpha \rightarrow 2$ limits of this article’s normal form, budget (the omitted intensity $n\beta a/(\alpha M^\alpha)$ tends to $1/(2s^2)$) and fold drift (Theorem 8.1). *Joint limit* $\alpha_n \rightarrow 2$: answered, again independently, by the confluent package [10], for $\varepsilon_n = 2 - \alpha_n \rightarrow 0$ at any rate and of either sign, at leading order, and by the stable–Gaussian package [11], in the window $|2 - \alpha_n| \log n \leq K$, one order further (a two-term uniform inverse, the first coefficient correction, and a uniform first cutoff correction without finite-prefix moments). Where both apply they agree term by term, and at $\alpha = 2$ they reduce to the formulas of the two fixed-endpoint packages. Partial progress at the lower endpoint, on the interior-fold side: the later lower-endpoint package [12] (batch 51) treats the pure tail $a_j = j^{-2-\varepsilon}$, $\alpha = 1 + \varepsilon \downarrow 1$, for couplings $\beta > \beta_c$ (a confluent normal form uniform through $\varepsilon = 0$, a Landau coefficient profile, the budget $M\delta \rightarrow \infty$ in its fold distance δ , which is not the δ_β of this article, and a doubly exponential cutoff loss). On the boundary-critical path itself, $a = 1$ and $\beta = \beta_c = 1/(\zeta(1 + \varepsilon) + P'(1))$ with the finite prefix P held fixed ($P = 0$ gives $\beta_c = 1/\zeta(1 + \varepsilon)$), the lower endpoint is answered, for $\varepsilon_n \rightarrow 0$ at any rate, by two independent later packages [13, 14] (batch 52), which prove the same theorem. The critical law (4.2) holds uniformly as $\alpha \downarrow 1$, in [14] with the relative error $O(\varepsilon n/b_n^2 + n^2 e^{-\eta n})$ and to every fixed order, where b_n is the scale (5.4) at $\beta = \beta_c$. Conditioning on the coefficient produces one exceptional action, whose deletion leaves the unconditioned configuration. When $n\varepsilon \rightarrow \infty$ the cutoff profile is a Landau law on the scale εb_n , centred at $b_n + \varepsilon b_n \log(1/\varepsilon)$; when $n\varepsilon$ stays bounded, n minus the largest action has a compound-Poisson limit. The constants of the two packages agree: their Landau variables differ by the constant $1 - \gamma$, with γ Euler’s constant, and their cutoff quantiles coincide. The subcritical paths $\beta < \beta_c$, couplings in the window (5.4) with $\lambda_n \neq 0$ as $\alpha \downarrow 1$, and a matching with the interior-fold chart of [12] stay open.

5. Nonlinear slopes that break the exact substitution. For $U = \beta \sum a_j q^j \exp((j+r_j)U)$ with $r_j = o(j)$, identify conditions under which the stable boundary and budget persist. The parametrization $t = qe^U$ is then no longer exact. A useful first case is $r_j = \gamma j^\theta$, $0 < \theta < 1$, with a controlled complex domain.

6. Incommensurate actions and shrinking gaps. Replace integer actions j by a genuinely incommensurate sequence, or let a family of action gaps tend to zero. Determine whether the critical inverse still admits a locally finite positive support semigroup and which effective summability conditions replace the lattice Fourier argument. This would address a different part of the repository’s parameter-dependent-action question.

7. Higher-order finite-fold drift. Expand $M \log \tau_M$, $\log(r_M/\rho_c)$, and v_M beyond their leading limits, allowing finite-prefix corrections. Establish a remainder strong enough to exponentiate after multiplication by n in a prescribed joint range. This would turn the warning after Theorem 8.2 into a usable sharp uniform formula.

Editorial note (ProveIt, 2026-09-29). Answered at the endpoint $\alpha = 2$ only. The logarithmic-endpoint package [8] gives the finite-fold drift to all inverse-logarithmic orders (its `thm:foldchart`) with an explicit condition for exponentiating it (its `prop:exponentiation`); the marginal package [9] gives a convergent reduced chart for the drift (its `thm:finite-fold`)

without an exponentiation criterion. For $1 < \alpha < 2$ the question is open.

8. Minimal support and deeper cancellations. The grid in (3.5) is sufficient, not always minimal. Classify finite prefixes which suppress $d_2, d_3, \dots$ and derive the resulting first nonzero sector corrections. When rational α causes exponent collisions, seek an effective criterion for cancellation of the entire grouped coefficient rather than only its gamma factor.

9. Divergent outer expansions and resurgence. Determine the large-order growth of the coefficients in Theorem 5.2, optimal truncation, and possible Borel or accelerated summation. Locate the relevant complex saddles and singular values. Convergence of the local Hahn expansion and the coefficient duality alone do not prove any of these claims.

10. Signed weights and competing complex singularities. Allow controlled negative or complex a_j and replace positivity by explicit continuation and saddle hypotheses. Determine when a multi-saddle analogue of the cutoff profile exists. The nonnegative Poisson conditioning proof will not survive unchanged, so an analytic replacement is necessary.

11. Proof-producing computation. Implement a checker for finite support bounds, exact rational coefficient identities, and local holomorphic majorants, followed by an interval Fourier integrator. Connect its certificates to Lean statements with separate hypotheses for analytic continuation, aperiodicity, and tail bounds. The deliverable should distinguish a verified coefficient from a verified asymptotic error estimate.

12 Conclusion

The decisive distinction is not merely between finitely many and infinitely many formal symbols. It is whether the action sum remains analytic at the boundary reached by the inverse. In the positive power-tail class, the failure of a finite second moment changes the critical exponent from $1/2$ to $1/\alpha$. Nevertheless, the inverse still possesses a convergent, explicitly computable Hahn chart with controlled support.

The same chart determines the all-order sector coefficients through an exact gamma reflection identity. It explains why some apparently leading corrections vanish, and why finite ramification is an arithmetic property of α . Truncating the countable action sum restores ordinary folds, but preserving large sectors requires a sharp growing action budget. The joint density profile supplies the missing transition, and the finite-fold drift gives its complementary analytic explanation.

These results provide a rigorous model in which local inversion, large-sector asymptotics, and action truncation can be compared without interchanging limits informally. The broader questions of arbitrary transseries realization, general resurgence, and noncommensurate action families remain distinct research tasks.

A Source audit and proof dependencies

A.1 Repository snapshot and limits of inspection

The repository sources used for the mathematical crosswalk were read at commit

a795fcffa4b3ece22761d81fbed3c43be8b81795.

The inspected group is

Analysis/Transseries/docs/series-and-transseries/.

The group README’s arrival descriptions and status caveats, the regularity package README, and the moving-fold article’s final research-question and scope sections were inspected directly. The full canonical transseries volume was not audited. In particular, the comparison above is not an assertion that every possible overlap in that volume has been excluded.

During inspection, `main` advanced to

5804c7aff9955e6cbc09be2a71a295a07b238bc9,

whose parent is the pinned revision above. That later commit records a new incoming-archive batch. This article does not claim to have reviewed those new archives. The repository was not modified.

The group README describes the recent research packages as unmerged, not claim-by-claim reviewed, and not Lean-formalized. Those qualifications are preserved here rather than treating their statements as certified premises. The underlying model and every theorem in this article are proved directly.

A.2 Dependency structure

The analytic branch is

polylogarithm expansion $\implies$ exact normal form $\implies$
convergent Hahn chart and coefficients.

The sector branch is

Lagrange identity $\implies$ Poisson coefficient representation $\implies$
Fourier local limits and corrections.

Gamma reflection identifies the coefficients from these two branches. The cutoff branch adds only tail Riemann sums, uniform Fourier bounds, and the exact conditional identity. The finite-fold branch uses a positive real root equation, its Riemann-sum limits, and a tilted Gaussian local limit. Numerical verification is downstream of all four branches; none uses numerical observations as a theorem hypothesis.

B A compact notation ledger

Symbol	Meaning
α, a, P	Tail exponent parameter, tail amplitude, and finite-prefix polynomial in $A(t) = a \operatorname{Li}_{1+\alpha}(t) + P(t)$.
d_k	Analytic coefficients in (2.2); only d_0, d_1 are ordinary finite moments.
G	$\Gamma(-\alpha) > 0$.
β_c, δ_β	Critical coupling $1/d_1$ and linear defect $1 - \beta d_1$.

Symbol	Meaning
ρ_β, B_β	Boundary value $e^{-\beta d_0}$ and fractional coefficient βaG .
x, ε	$-\log t$ and $\log(\rho_\beta/q)$ near the boundary.
T, R, V	Hahn chart variables $T = (\varepsilon/B_c)^{1/\alpha}$, $R = T^{2-\alpha}$, $x = TV(T, R)$.
$\mathbf{m}, m_*, K, \Lambda$	Multi-index, its total multiplicity, weighted index, and Hahn exponent numerator in (3.7).
b_n, λ_n	Stable sector scale and centered critical-window coordinate in (5.4).
$g_\alpha, g_{\alpha,L}$	Full and upper-jump-truncated centered stable densities.
ν_L, h_L	Omitted-jump intensity and omitted mean at cutoff L .
$\mathcal{R}_\alpha(L, \lambda)$	Limiting retained-coefficient fraction in (7.5).
τ_M, r_M, v_M	Exact finite-action critical point, critical value, and tilted curvature.
$v_\alpha, D_\alpha, V_\alpha$	Limiting finite-fold drift constants; v_α is distinct from the curvature v_M .

References

- [1] NIST Digital Library of Mathematical Functions. Section 25.12, *Polylogarithms*, especially Eq. 25.12.12. <https://dlmf.nist.gov/25.12>. Consulted 29 September 2026.
- [2] P. Flajolet and R. Sedgewick. *Analytic Combinatorics*. Cambridge University Press, 2009. Author-maintained book site: <https://ac.cs.princeton.edu/>.
- [3] S. Janson. Simply generated trees, conditioned Galton–Watson trees, random allocations and condensation. *Probability Surveys* **9** (2012), 103–252. doi:10.1214/11-PS188. Preprint [arXiv:1112.0510](https://arxiv.org/abs/1112.0510); see the critical infinite-variance discussion and Example 18.27.
- [4] A. Chakrabarty and G. Samorodnitsky. Understanding heavy tails in a bounded world or, is a truncated heavy tail heavy or not? *Stochastic Models* **28**(1) (2012), 109–143. doi:10.1080/15326349.2012.646551. Preprint [arXiv:1001.3218](https://arxiv.org/abs/1001.3218).
- [5] VladimirReshetnikov/ProveIt repository. *Series and transseries*, group README and arrival-status descriptions. Path `Analysis/Transseries/docs/series-and-transseries/README.md`. Snapshot and inspected scope: Appendix A. <https://github.com/VladimirReshetnikov/ProveIt>.
- [6] VladimirReshetnikov/ProveIt repository, research manuscript. *Critical Transseries at a Moving Fold: a uniform two-sheet atlas, large-sector crossover, and resonant action splitting*. Path below the preceding group: `Critical_Transseries_Moving_Fold/article.tex`. Final research-question and scope sections, source lines 1400–1580, read at the pinned snapshot. September 2026.
- [7] VladimirReshetnikov/ProveIt repository, research package. *A Sharp Regularity Classification for Countable Exponential-Feedback Transseries*. Path below the group: `Exponential_Feedback_Regularity_Classification/README.md`. Package model, verification status, and limitations read at the pinned snapshot. September 2026.
- [8] [Editorial addition, ProveIt, 2026-09-29.] *The Logarithmic Critical Endpoint: Convergent Lambert Charts, All-Order Sector Laws, and Sharp Action-Cutoff Corrections*, research package filed in batch 48. Path below the group: `Logarithmic_Critical_Endpoint_Lambert_Charts/article.tex`.
- [9] [Editorial addition, ProveIt, 2026-09-29.] *Marginal Critical Transseries: Convergent Lambert-W Charts, Universal Cutoff Corrections, and Certified Action Budgets*, research package filed in batch 48. Path below the group: `Marginal_Critical_Transseries_Action_Budgets/Marginal_Critical_Transseries.tex`.
- [10] [Editorial addition, ProveIt, 2026-09-29.] *Confluent Critical Transseries Across the Exponent-Two Boundary*, research package filed in batch 49. Path below the group: `Confluent_Critical_Transseries_Exponent_Two_Boundary/Confluent_Critical_Transseries.tex`.

- [11] [Editorial addition, ProveIt, 2026-09-29.] *Through the Stable-Gaussian Endpoint: Uniform Critical Transseries, Prefix-Independent Cutoff Corrections, and Conditional Extremes*, research package filed in batch 49. Path below the group: `Stable_Gaussian_Endpoint_Uniform_Critical_Transseries/article.tex`.
- [12] [Editorial addition, ProveIt, 2026-09-29.] *Through the Lower Critical Endpoint: Exponentially Coalescing Folds, Landau Transseries, and Conditional Action Budgets*, research package filed in batch 51. Path below the group: `Lower_Critical_Endpoint_Landau_Transseries/article.tex`.
- [13] [Editorial addition, ProveIt, 2026-09-29.] *The Lower Critical Endpoint: Arbitrary-Rate Condensation, Cauchy Cutoff Laws, and Degenerating Transseries*, research package filed in batch 52. Path below the group: `Lower_Critical_Endpoint_Cauchy_Cutoff_Condensation/lower_critical_endpoint.tex`.
- [14] [Editorial addition, ProveIt, 2026-09-29.] *The Lower Critical Endpoint of Exponential-Feedback Transseries: Uniform Coefficient Asymptotics, Deletion of the Largest Action, Landau Cutoff Profiles, and the Compound-Poisson Boundary*, research package filed in batch 52. Path below the group: `Lower_Critical_Endpoint_Uniform_Coefficients_Compound_Poisson/article.tex`.